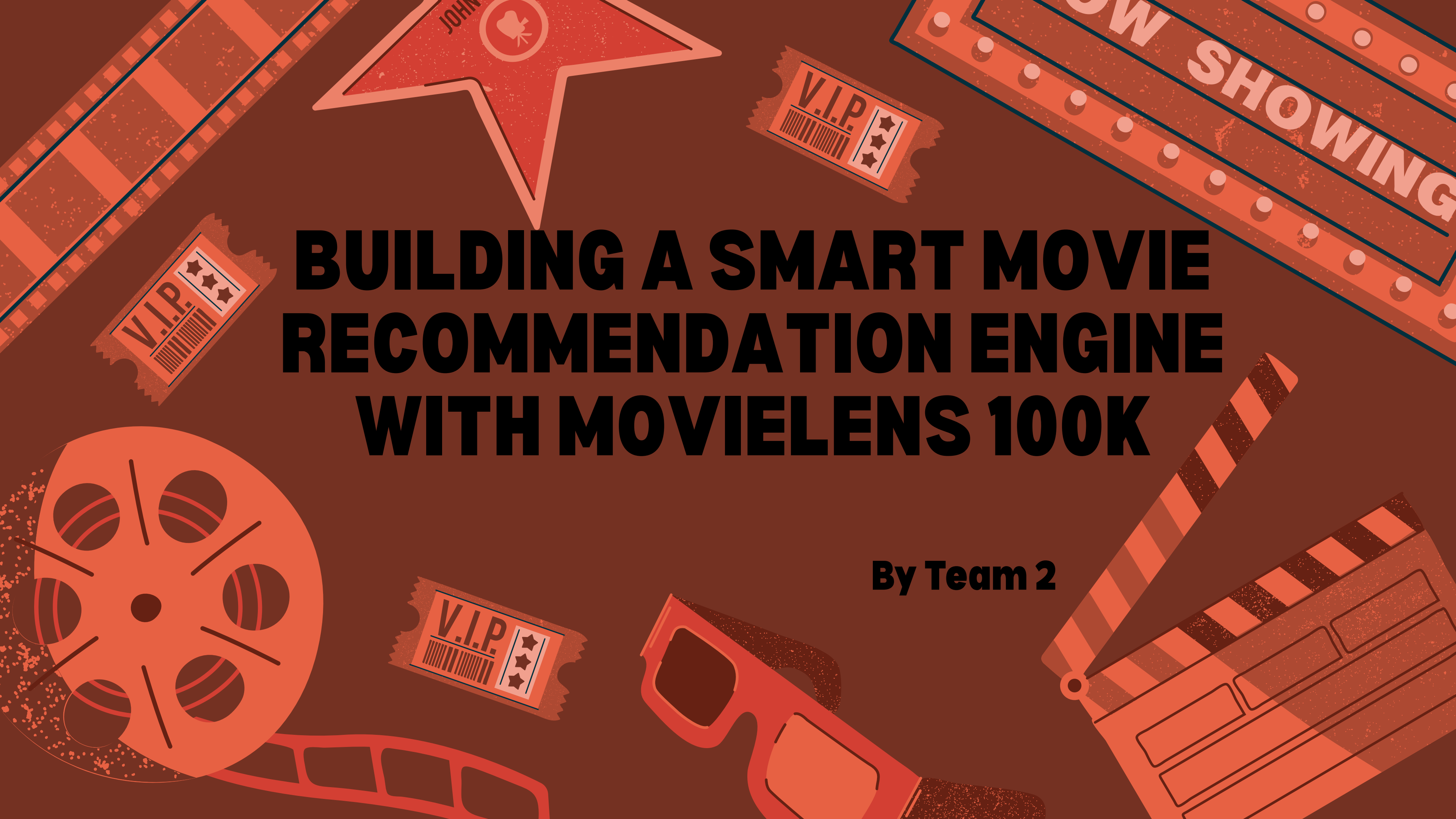
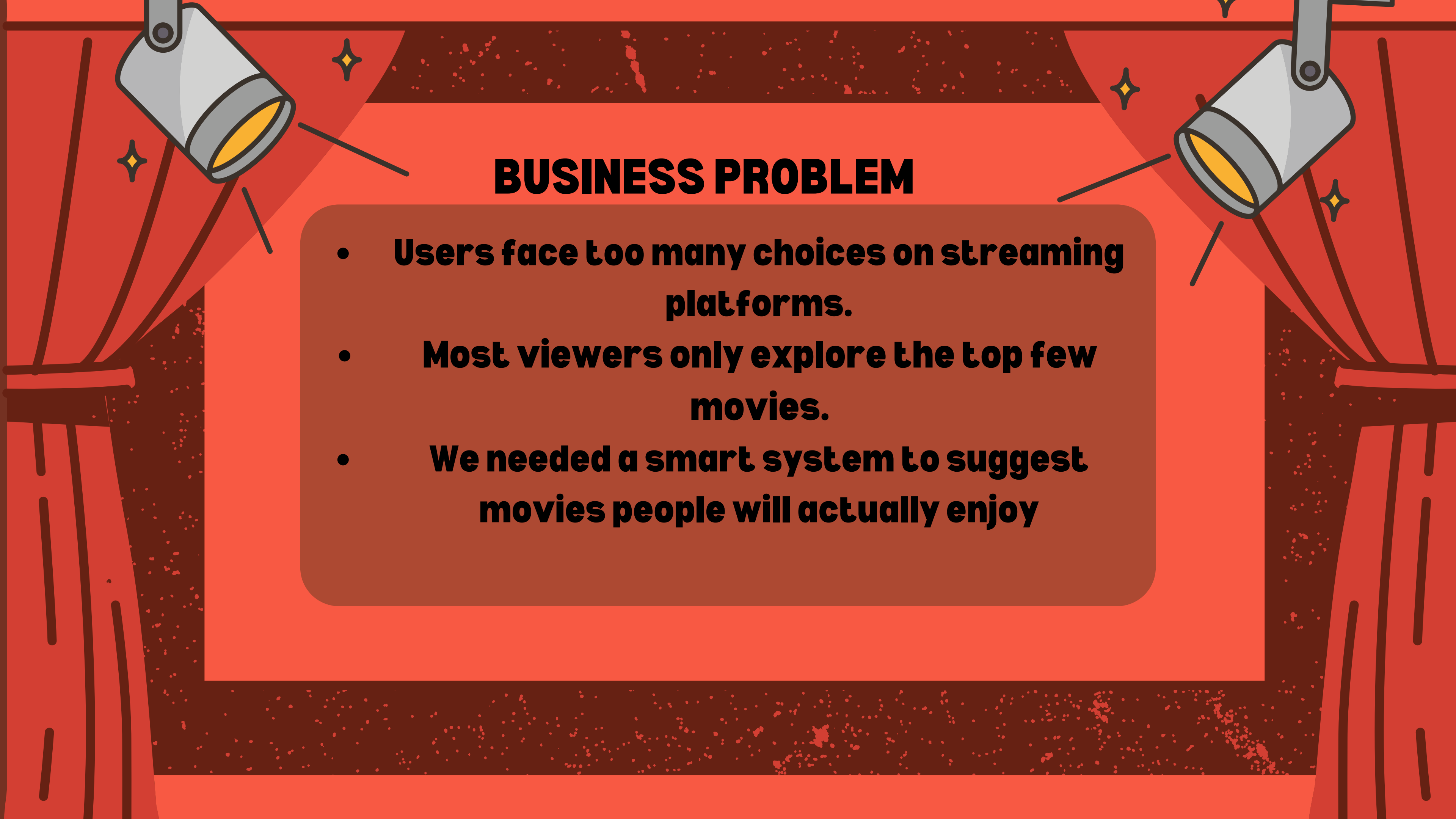


A collage of movie-related items in shades of orange and red, including a film strip, a star with 'JOHN' and a popcorn icon, a 'V.I.P.' ticket, a 'NOW SHOWING' marquee, a film reel, a clapperboard, and 3D glasses.

BUILDING A SMART MOVIE RECOMMENDATION ENGINE WITH MOVIELENS 100K

By Team 2

The background of the slide is a stylized illustration of a stage. It features red curtains on the left and right sides, held back by silver rings. Two spotlights are mounted on the top of the stage, casting a warm yellow glow. The stage floor is a dark brown color with a speckled texture. The overall aesthetic is clean and modern, with a focus on the central text.

BUSINESS PROBLEM

- **Users face too many choices on streaming platforms.**
- **Most viewers only explore the top few movies.**
- **We needed a smart system to suggest movies people will actually enjoy**

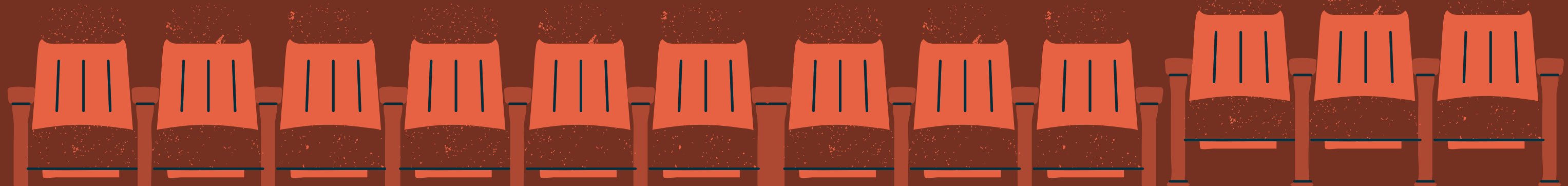
OBJECTIVES

- To offer a personalized user experience.

- To promote product consumption by users to increase revenue.

- To reduce user fatigue when searching for a movie by creating a recommendation system to suggest top 5 new movies for a user based on their rating history

- To create a recommendation system to suggest five movies for a new user using content based filtering by factoring in the desired genre.



Project Overview

Sequence

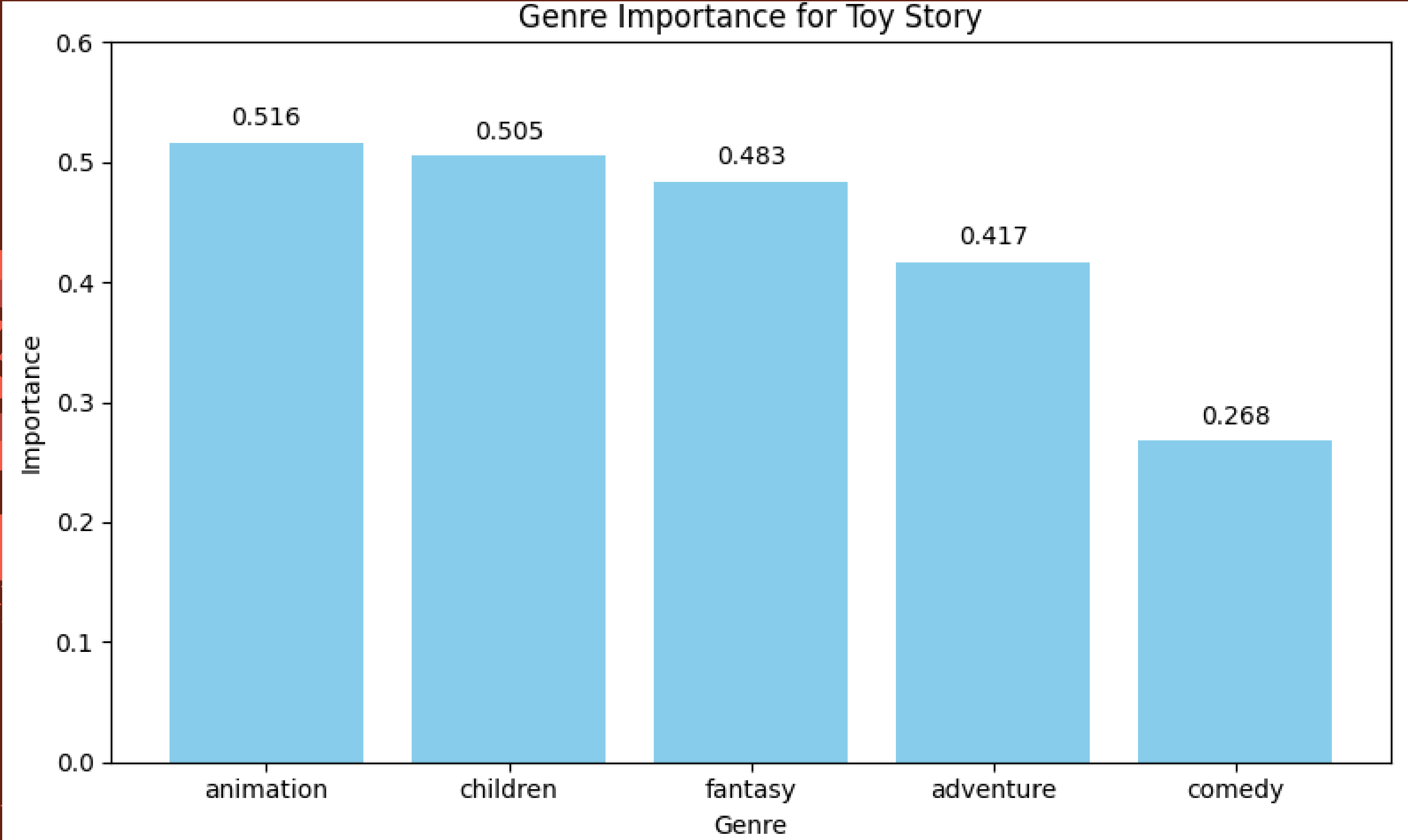
Dataset → Data Prep →
Content-Based Model →
Collaborative Filtering
→ Hybrid Recommender
→ Evaluation



About the Model:

- 100K ratings across 9,000 movies
- Combines content and collaborative filtering
- Tailors Top-5 recommendations per user

Feature importance analysis for content-based system



Using Toy
Story
example

Hybrid Recommender system(example for user 1)

Title		Genres	Predicted Rating	Hybrid Score	
Galaxy of Terror (Quest) (1981)		Action, Horror, Mystery, Sci-Fi	4.62	0.89	
The Big Bus (1976)		Action, Comedy	4.56	0.88	
The Assignment (1997)		Action, Thriller	4.51	0.87	
Last Hurrah for Chivalry (Hao xia) (1979)		Action, Drama	4.41	0.86	
Light Years (Gandahar) (1988)		Adventure, Animation, Fantasy, Sci-Fi	4.38	0.85	

The hybrid recommendation system combines both content-based and collaborative filtering approaches to leverage their respective strengths as shown in the diagram above.





Model Comparison and Results



Model	RMSE	MAE	Coverage	Precision@5	Recall@5
SVD	0.8807	0.6766	1.0	0.7550	0.3812
SVD_tuned	0.8807	0.6766	1.0	0.7550	0.3812
NMF	0.9288	0.7105	1.0	0.7441	0.3684
KNN_User	0.9077	0.6933	1.0	0.7174	0.3724
KNN_Item	0.9129	0.6953	1.0	0.7560	0.3870
SVD_Best	0.8637	0.6620	1.0	0.7717	0.3946

SVD_Best outperforms all others with the lowest RMSE (0.8604) and MAE (0.0064), indicating more accurate predictions and minimal error making it suitable for deployment in a real world scenario.



Business Impact

The Recommendation system demonstrates:

- Average Rating Improvement: 0.5018
- Improvement Rate: 0.7914
- High Confidence Predictions: 4780.0000
- Potential Engagement Boost: 0.2370
- Average Recommendation Diversity: 0.9200

Model Limitations

1. Data Sparsity: User-item matrix is highly sparse (>99% zeros)
2. Cold Start: Difficulty recommending to new users with no rating history
3. Popularity Bias: Tendency to recommend popular movies
4. Scalability: Real-time recommendations may face latency issues
5. Feature Engineering: Limited movie features beyond genres
bit of body text

Future Improvements

1. Implement deep learning models (Neural Collaborative Filtering)
 2. Add implicit feedback data (views, clicks, time spent)
 3. Incorporate user demographics and movie metadata
 4. Implement online learning for real-time model updates
5. Add diversity and fairness constraints to recommendations
6. Use ensemble methods combining multiple algorithms

A stylized illustration of a stage. At the top, a dark red curtain is partially open, revealing a bright red background. Two spotlights, one on the left and one on the right, are mounted on stands and pointed towards the center. The spotlights have grey bodies and yellow lenses. Several small, four-pointed yellow stars are scattered around the spotlights. In the foreground, a row of orange chairs with dark brown seats and backs is visible. The chairs have vertical lines on their backs. The overall color scheme is dominated by reds and oranges, with grey and yellow accents.

THANKYOU !